

Getting Started with SKAI3 LV

2023-01-19 Michael Uhl



Disclaimer

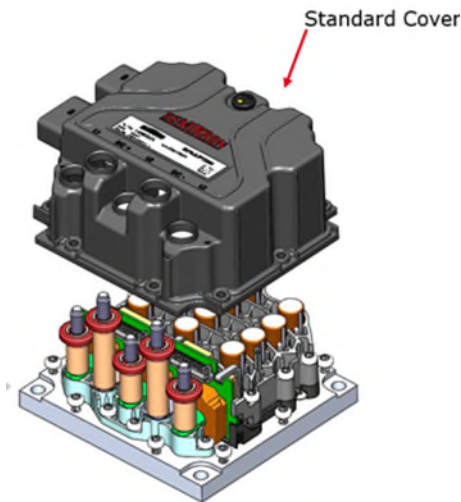
This presentation and included examples do not serve as replacement of the available detailed documentation for SKAI3 LV hardware.

Described mechanical and electrical details use the OLEA controller board (designed by Silicon Mobility) in an exemplary manner and may be insufficient depending on both target application and target controller board.

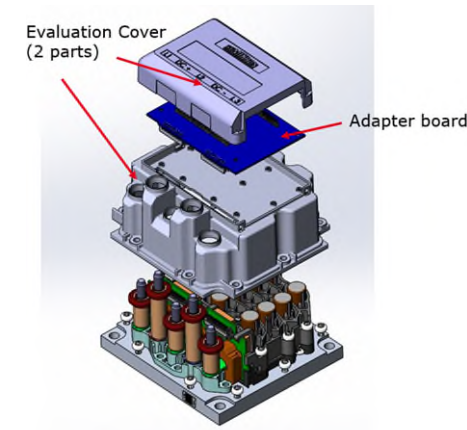
Responsibility of meeting standards, which are mandatory for the target application, remains with the customer.

SKAI3 LV Standard vs. Evaluation Kit Version

- The SKAI3 LV standard provides a defined space for the customer-designed controller board; see [\[1\] Overview Mechanical Dimensions](#)
- A controller board is not included in the SKAI3 LV
- → This "Getting Started" presentation focuses on the SKAI3 LV standard version

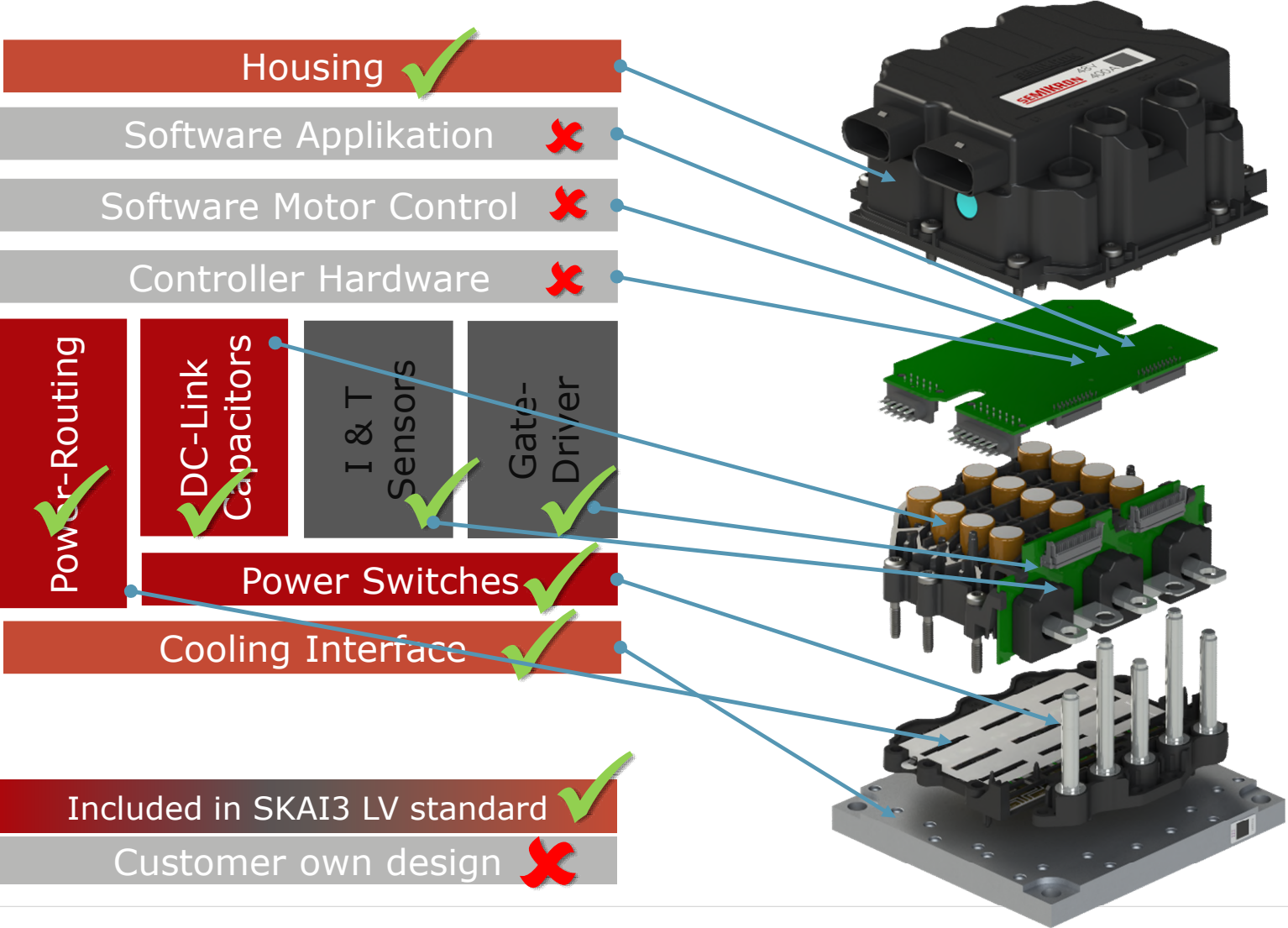


- For straight-forward evaluation purposes, the SKAI3 LV is also available as Evaluation Kit version; with the included adapter board, the Evaluation Kit version provides various electrical connection options avoiding the need for special connectors
- → A separate "Getting Started" presentation for the SKAI3 LV system as "evaluation kit" version is in preparation.



- For more details on differences of SKAI3 LV Standard vs. Evaluation Kit Version see [\[4\] SKAI3LV_ProductOverview](#)

SKAI3 LV Building Block Concept



Application-specific Requirements/Considerations

For appropriate use of the SKAI3 LV in the target application, various considerations need to be done

The list below provides an overview of the different questions with hyperlinks to specific details

- **IP protection?**
 - Refer to [Ingress Protection Requirements](#)
- **Installation?**
 - Refer to [Mechanical/electrical Installation Concept](#)
- **Cooler type?**
 - Refer to [Cooling Concepts](#)
- **Cooling performance requirements?**
 - Refer to [Cooler Selection](#)
- **(Electric) ground concept?**
 - For choice of suitable ground concept certain requirements may need to be considered
 - Refer to [Decision Matrix – Ground Concept](#)

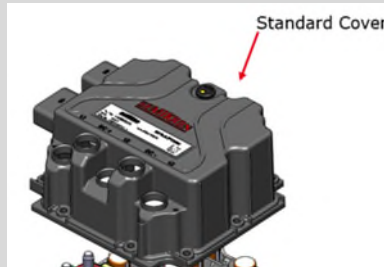
Ingress Protection Requirements

Ensure that provided ingress protection of the SKAI3 LV meets with application requirements.

SKAI3 LV Standard provides ingress protection up to IP 66 ^{*)} according to ISO 20653:2013.

IP 66 is given if

- the standard cover ,



- the IP seals ^{**)},



- and both vehicle signal connectors
are mounted.

^{*)} please note the IP 00 rating for the power terminals (i.e. no protection)

^{**)} for details on assembly of IP seals refer to [Annex D - Mounting IP seals on power terminals](#)

Mechanical/electrical Installation Concept

The requirements on the mechanical/electrical installation need to consider

- application-specific mechanical stress on SKAI3 LV (shock & vibration)
 - customer-specific standards applicable?
 - legal standards applicable?
 - for general information refer to Mechanical Installation of SKAI3 LV Standard
- application-specific assembly of power cables connected to SKAI3 LV
 - mechanical stress relief
 - thermal stress
 - for general information refer to Electrical Installation of SKAI3 LV Standard
- realisation of cooling requirements
 - refer to Cooling Concepts and subsequent slides

Cooling Concepts

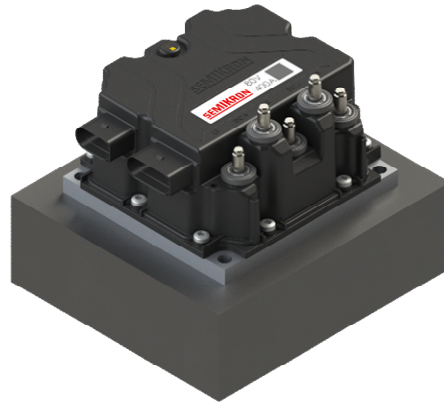
Cooling of SKAI3 LV standard

Heat transfer via cooling plate to vehicle structure or dedicated vehicle cooling-system

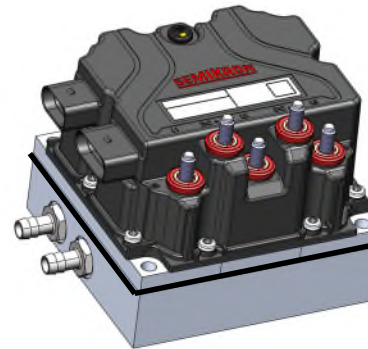
Integrated air or watercooler on request



Exemplary air cooler for heat transfer
(required fan and air ducts not displayed)



Exemplary metal block as
heat drain

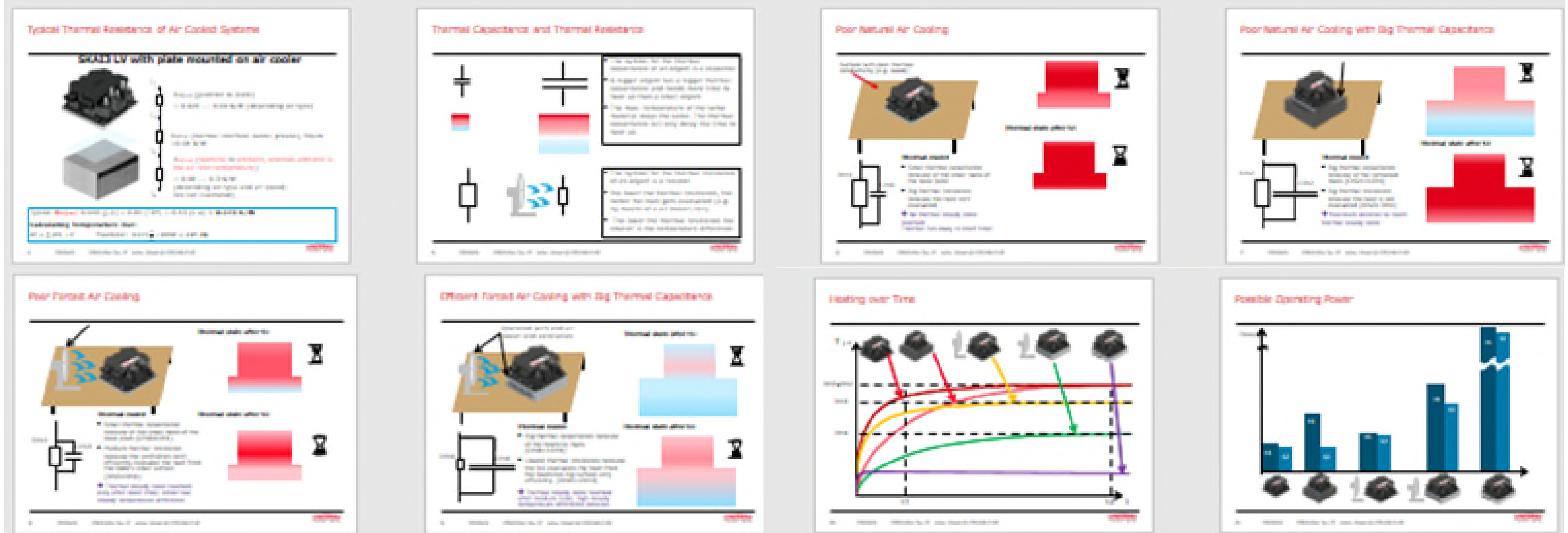


Exemplary water cooler for
heat transfer



...Continued on next slides...

Cooling Concepts



For a general comparison/characteristics of different cooling options see [2]
[SKAI3LV_ComparisonCoolingOptions](#)

Cooler Selection

To select a cooler optimised for the application, the power losses profile generated by the SKAI3 LV must be known.

The actual power losses to be transferred via the cooler mainly depend on parameters as follows:

- DC link voltage
- Output current (instantaneous value and RMS value)
- PWM switching frequency
- Operating junction temperature

These power losses consist of

- „on losses“ or „conducting losses“ generated during on-state of the power switch

plus

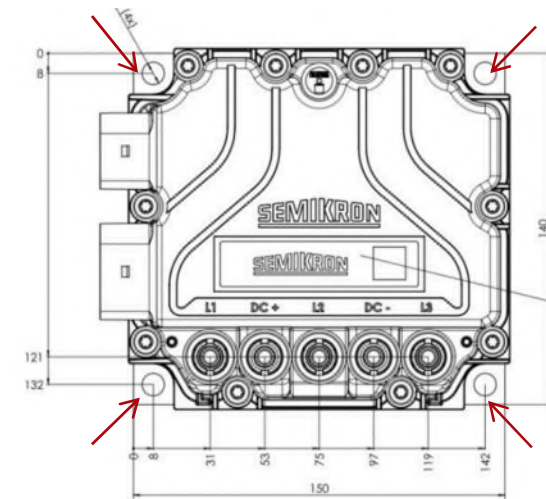
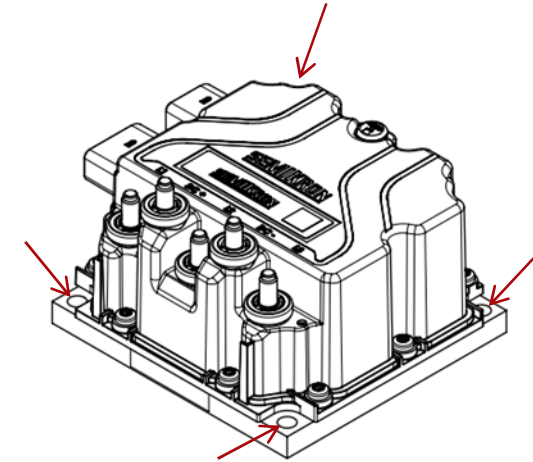
- „switching losses“ which are generated during transition between off-state and on-state of the power switch (and vice versa)

For calculation of the power losses in the desired application, please contact SEMIKRON

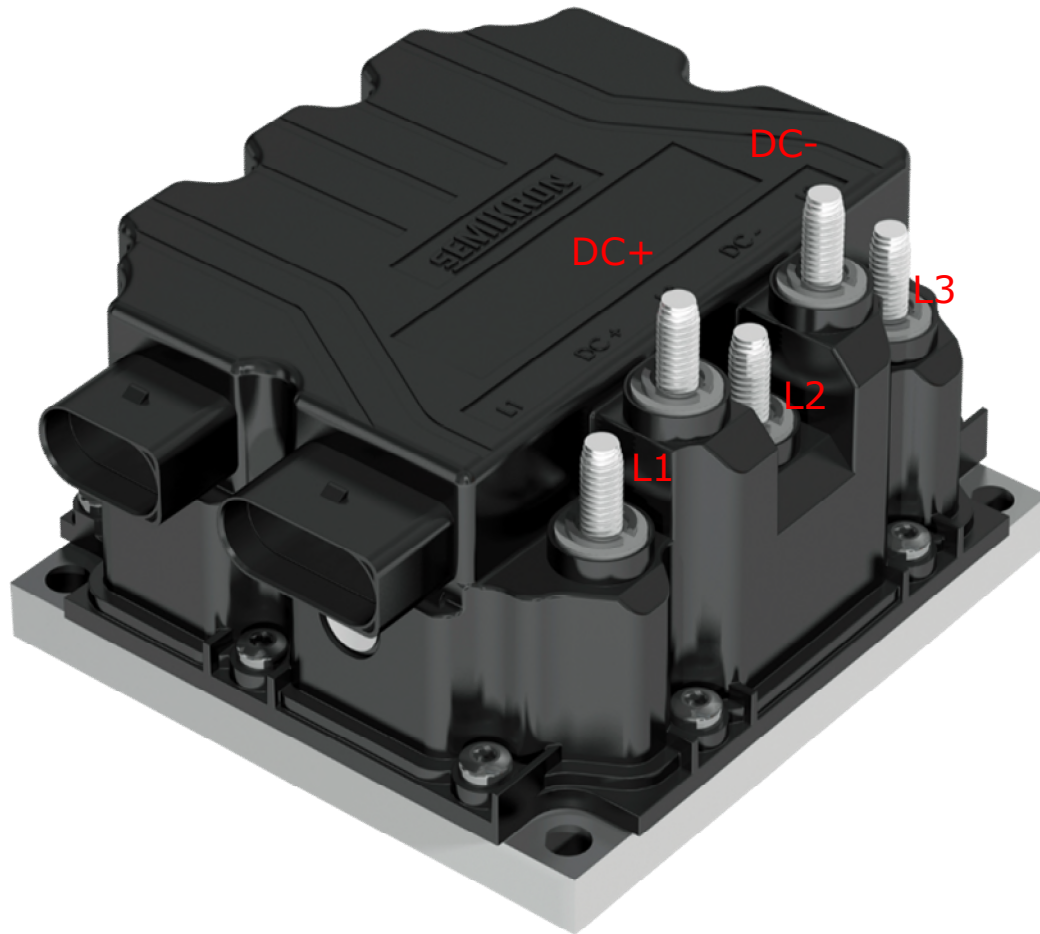
Mechanical Installation of SKAI3 LV Standard

Mounting the SKAI3 LV plate to Cooler

- Mounting of the cooling plate serves as mechanical fixation and thermal interface
- The ground plate must be installed on a **suitable cooler**; see next two slides Cooling Concepts for cooling requirements
- To ensure the required heat transfer from the plate to the mounting surface, a suitable thermal interface material (TIM) is required, and certain conditions apply to the mounting surface
- Main mounting holes for four M8 screws on the plate
- See [1] Mounting Recommendations of Cooling Plate for details.



Electrical Installation of SKAI3 LV Standard

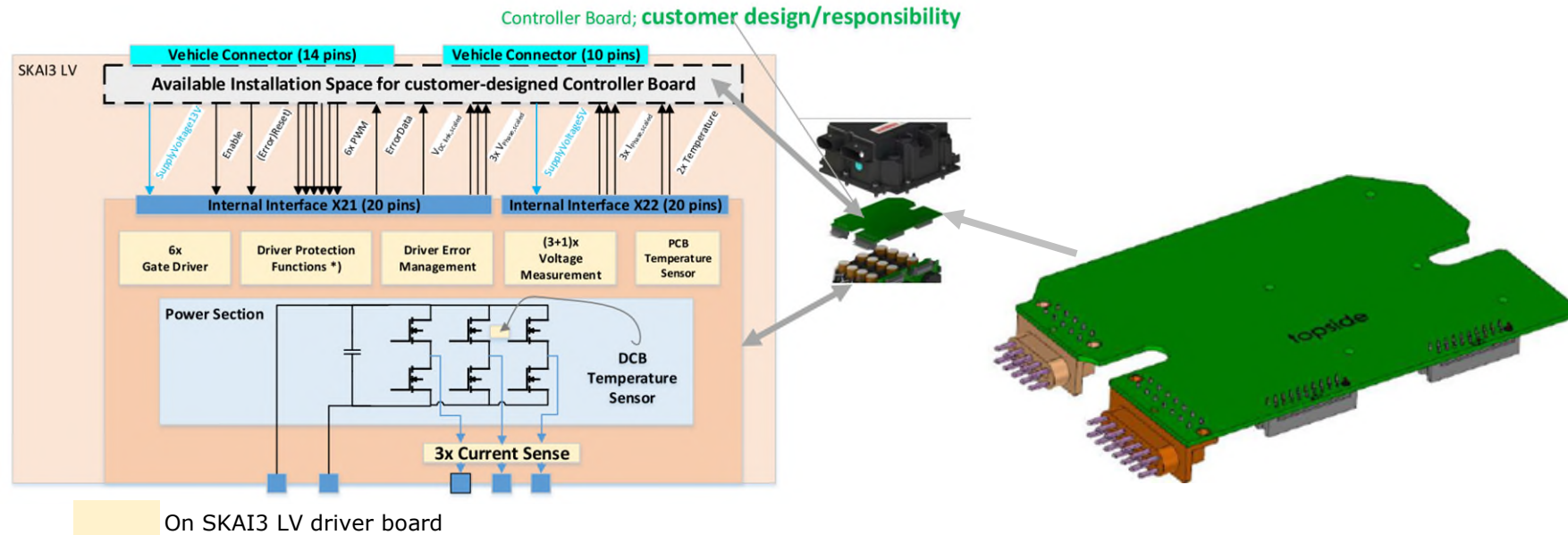


- Power Connections
- Electrical power connections via cable lugs
 - DC cable (from DC power source/battery)
 - AC cable (to electrical machine)
- The cable lugs must be fixed with a nut, a washer and a spring washer.
For required torque and maximum distance between cable fixture and power terminals see [1] Power Interface.
- **The reference potential of the customer controller board (control voltage GND) must be connected with the negative DC voltage potential!**
Refer to [1] Power Supply

Decision Matrix – Ground Concept

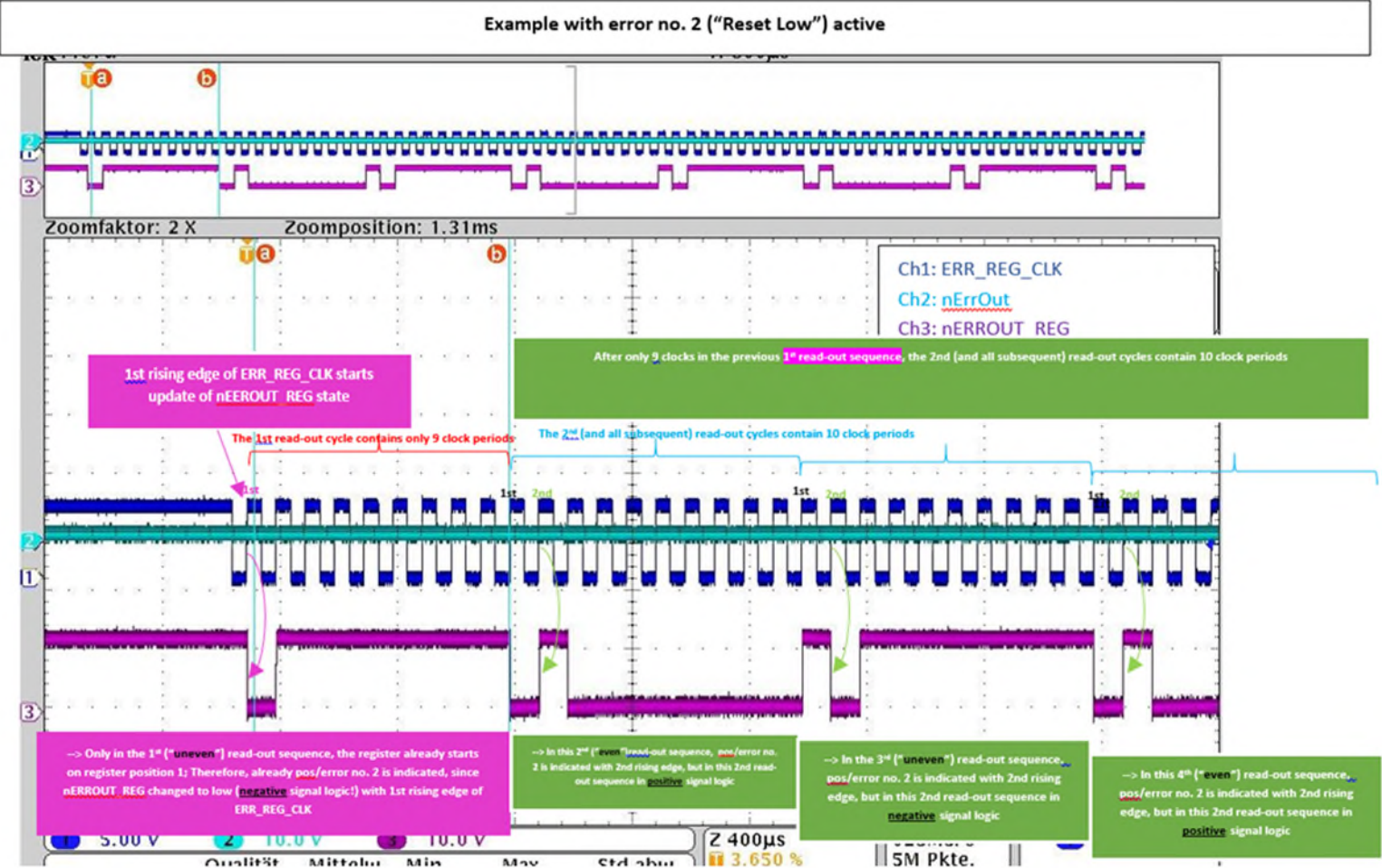
	Ground concept 1...6 → Refer to [1] Ground Concept					
	1	2	3	4	5	6
Suitable for voltage class	SKAI XX XX MD 08-X	SKAI XX XX MD 08-X SKAI XX XX MD 10-X SKAI XX XX MD 15-X				
Absolute maximum battery voltage	≤60 V	≤60 V ≤72 V ≤120 V				
Electrical connection of DC-to chassis?	Optional	Optional possible for ≤60 V Not allowed for >60 V				
GND-to-DCminus connection	Internal on controller board only					External via vehicle connector only
Pro / Con	Non-isolated voltage supply possible for SKAI3 LV and controller board	Requires isolated voltage supply possible for SKAI3 LV and controller board and isolated communication to VCU	Non-isolated voltage supply possible for SKAI3 LV and controller board; Requires isolated communication to VCU	Requires isolated voltage supply possible for SKAI3 LV and controller board and isolated communication to VCU	Requires complete electrical isolation on internal X21 interface between controller board and SKAI3 LV driver	Non-isolated voltage supply possible for SKAI3 LV and controller board Mandatory GND-to-DCminus connection needs to be established during individual system installation process

Customer-designed Controller Board



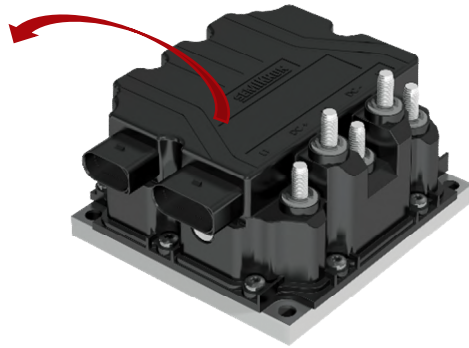
- The SKAI3 LV provides an internal installation space for a customer-designed controller board.
- → For physical dimensions/constraints refer to [1] Customer-designed Controller Board for SKAI3 LV Standard Version
- → For electrical interface data between SKAI3 LV driver board and customer-designed controller board see [1] Control interface of SKAI3 LV

Exemplary Read-out of Error Type

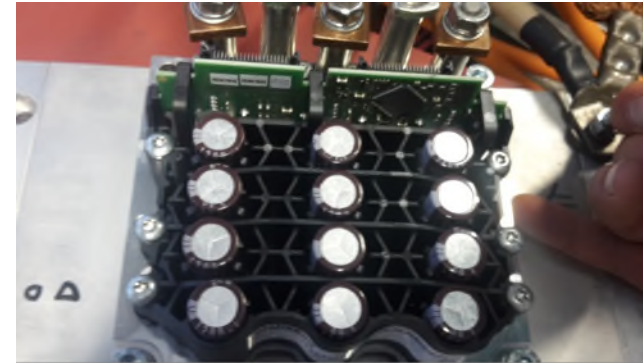


Assembly Steps of a Controller Board to SKAI3 LV

- Disassembly of standard cover



SKAI3 LV in opened state



Electrical connection and assembly of controller board to SKAI3 LV

In a series production approach, the controller board is first mounted to the inside of the removed standard cover.

This combination is then assembled to the SKAI3 LV. **For details see Annex C.**

Exemplary Installation of SKAI3 LV with OLEA Controller Board

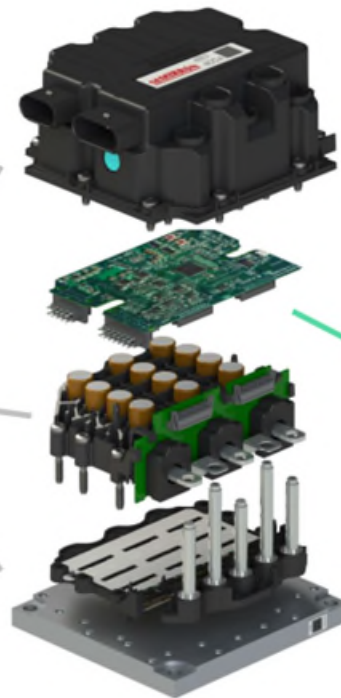
Exemplary Installation of SKAI3 LV with OLEA Controller Board

SKAI3LV-OLEA (SKAI3 LV with Silicon Mobility Controller Board)



SEMIKRON Advanced Integrated MOSFET Module

- SKAI3 LV is a MOSFET module with connected DC-link, gate-driver, protection functions and sensors
- SKAI3 LV is completed performance power-solution for 3-phase motor-drives
- SKAI3 LV reduces the time-to-market with the ready-made power-section



OLEA® Inverter Control Solution for SKAI3 LV Module

- OLEA® Inverter Solution includes 3 items :
 - I. A programmable control chip +
 - II. A control application software +
 - III. A control board reference design
- Embeds state-of-the-art control algorithms and safety functions
- Available in 2 levels of customization :
Ready to use or fully customizable

Assembly Steps

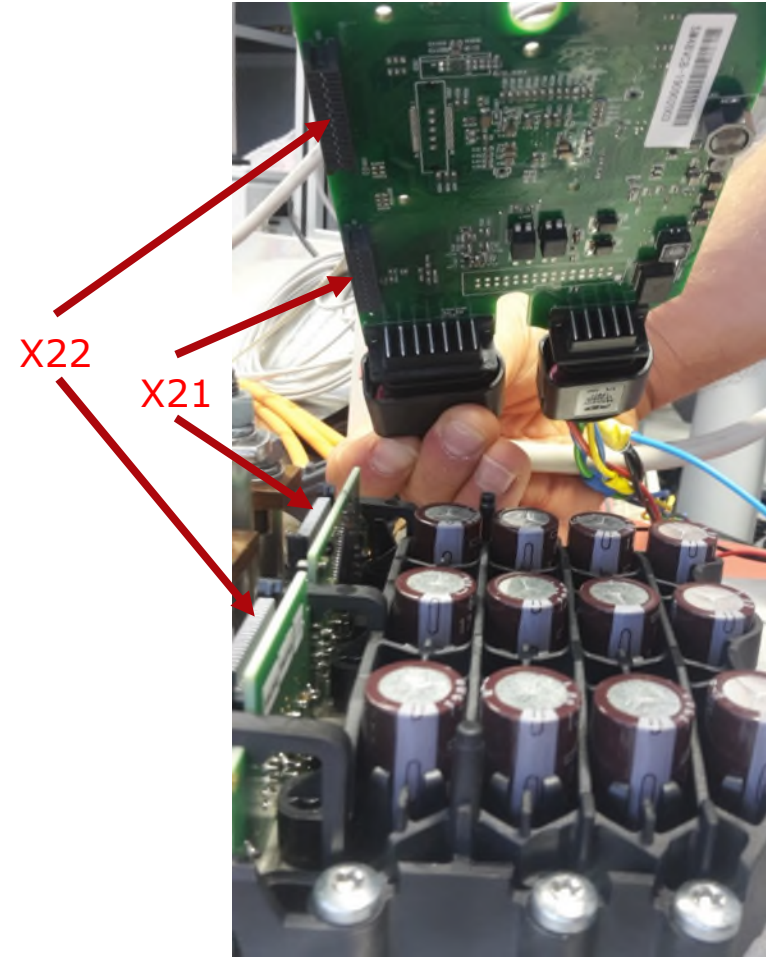
Controller Board to SKAI3 LV

Electrical connection of Controller board to SKAI3 LV

NOTE: Normally, the controller board would first be mounted to the inside of the removed standard cover. This combination would afterwards be assembled to the SKAI3 LV. For details see [Annex C](#).

For the commissioning as exemplarily described with this presentation however, the standard cover was not assembled anymore.

Thus the controller board was placed directly on the SKAI3 LV as shown here and the next slide.

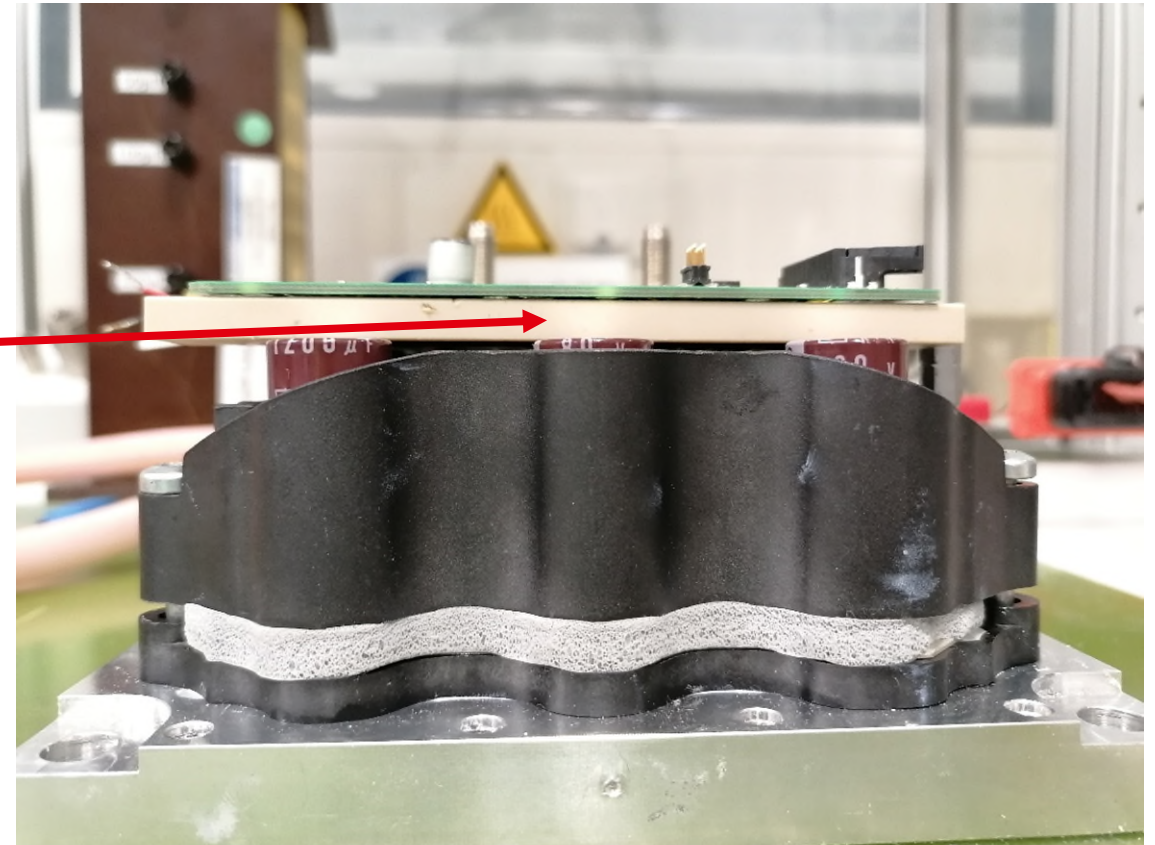


Assembly Steps of Controller Board to SKAI3 LV

CAUTION:

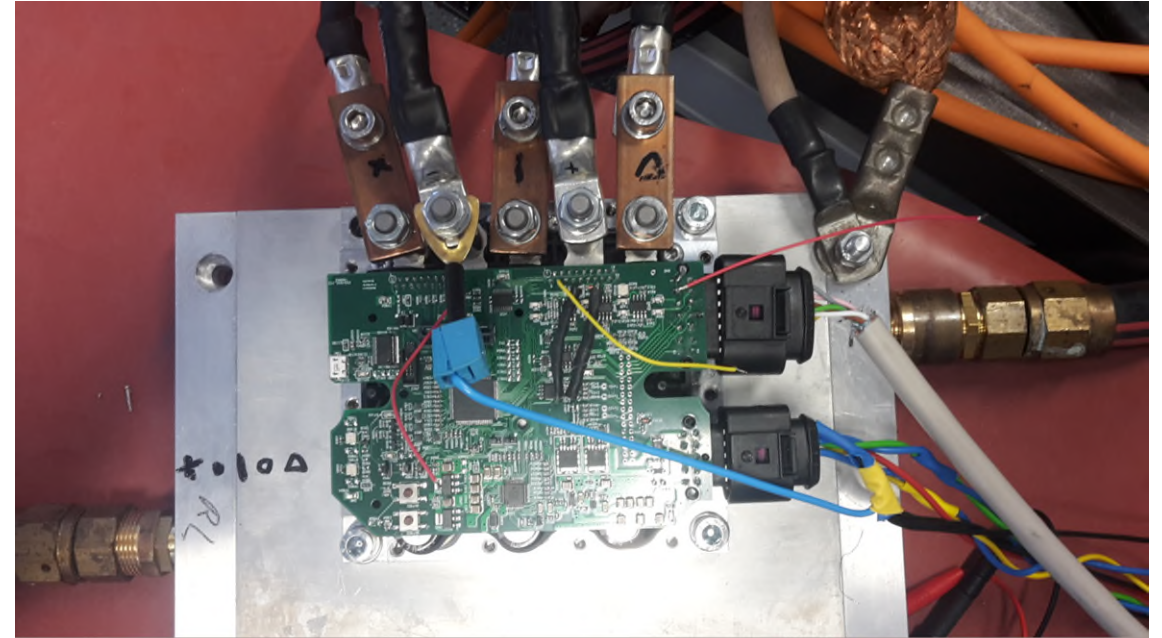
Conducting DC link capacitors surfaces are not fully electrically isolated from the dc link circuit.

When assembling the controller board without the SKAI3 LV standard cover, it has to be made sure to provide the electrical isolation between any PCB components and the dc link capacitors surfaces. This can be achieved by application of a suitable **insulation material**.

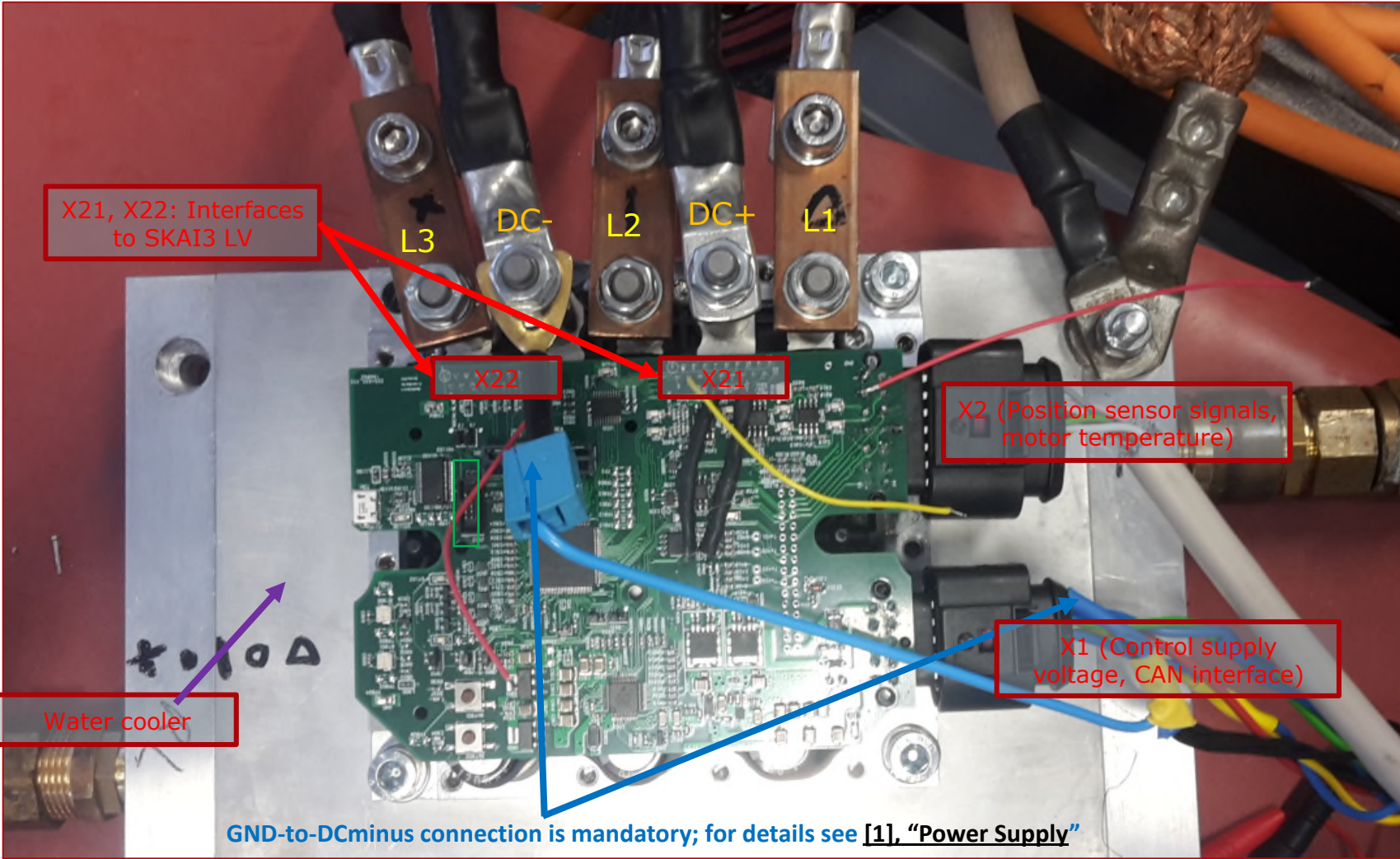


Controller Board assembled to SKAI3 LV

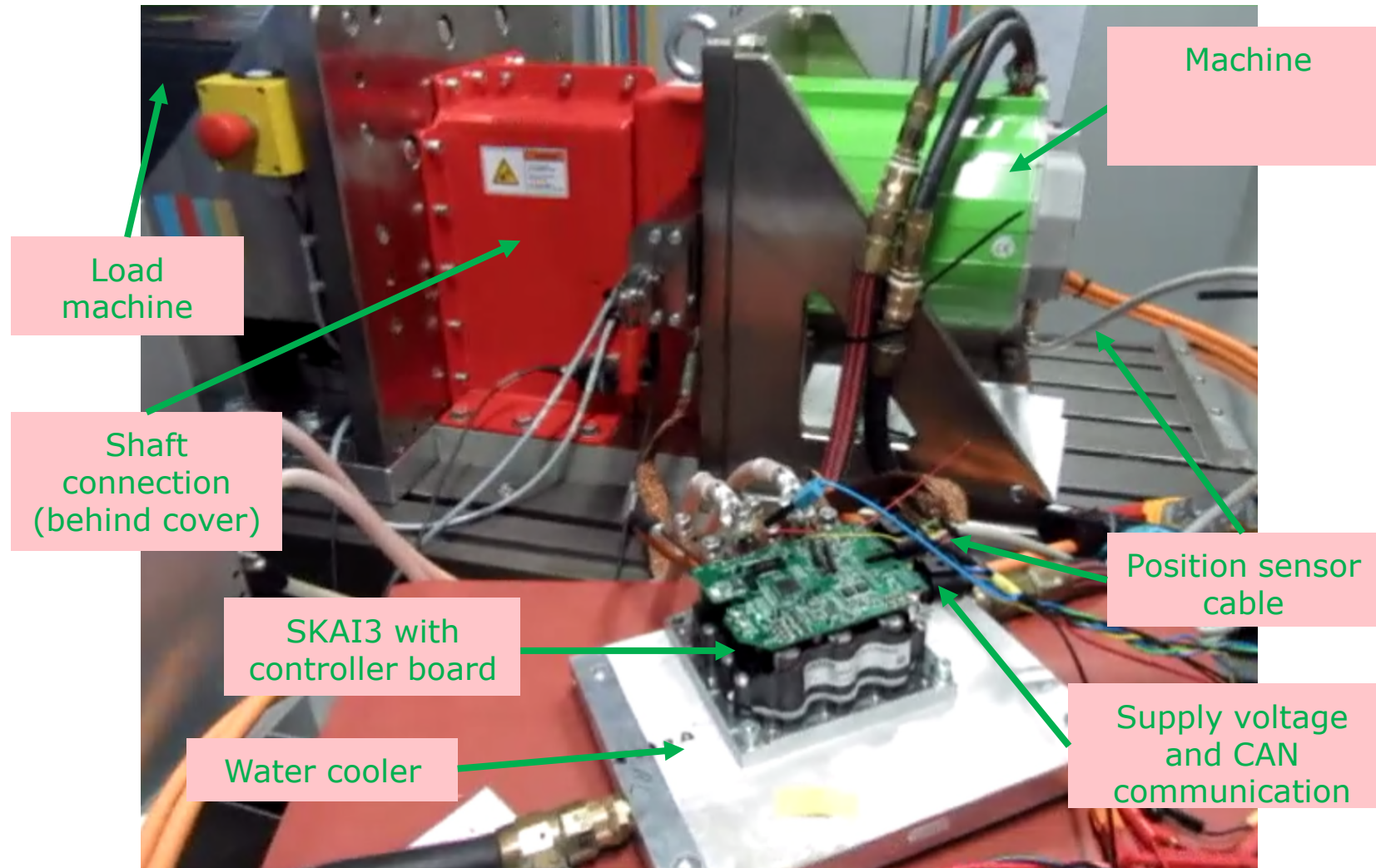
For the commissioning as described with this presentation, the standard cover was not assembled anymore. This was done in order to enable direct access to the programming interface on the controller board for persistent parametrisation updates during in the course of this commissioning.



Overview of Installed Controller Board on SKAI3 LV



Assembly Overview of SKAI3LV in Testbench Environment



References

- [1] SKAI3LV_technical_explanation_V2_2.pdf
- [2] SKAI3LV_ComparisonCoolingOptions_2019.10.27.pdf
- [3] SKAI3LV_eval-kit_adapter-board_V2_0.pdf
- [4] SKAI3LV_ProductOverview_2020.01.21.pdf

Annex A - Excerpt of SKAI3 LV Datasheet

SKAI 41 A3 MD 10 P Rev. 1.1



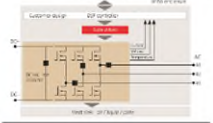
3-phase MOSFET inverter
14286010-K01

- Inverter Features**
Power-stage with:
- Gate-driver with protection functions
 - DC-link capacitors
 - 2 current-sensors
 - Temp. sensors
- Prepared for integration of controller board and software
- High power density
 - High overload capability

Environmental conditions:

- according to ISO 16750-D,E,K,L
- according to ISO 20653 IP6K6

- Typical Applications**
- Drives for battery-powered applications
 - Fork-lift trucks, mobile platforms & tow-trucks
 - Golf-carts
 - Light EV
 - Motor-bikes & quads
- not suitable for mains applications



Characteristics					
Symbol	Conditions	min.	typ.	max.	Unit
Absolute maximum ratings					
V_{in}^1	AC (sinus 50Hz), $t = 1s$			1500	V
V_{DC}	Maximum DC supply voltage / DC link voltage			80	V
$\Delta V_{GND-DClink}$	Maximum offset voltage between GND and DC-	-25		25	V
T_{jmax}	Maximum junction temperature T_j of power-switch			130	°C
$T_{amb,max}$	Maximum temperature T_{amb} of DC-link capacitors			100	°C
Electrical Data					
V_{CC}	DC supply voltage / DC link voltage	12	48 ²	72	V
Operating conditions for all output currents	$V_{DC}=48V$, $T_a=40^{\circ}C^1$				
	$f_{sw}=50\text{ Hz}$, $f_m=8\text{ kHz}$				
	$R_{thJA}=0.17\text{ KW}^{\circ}C$, $\cos(\phi)=0.8$, m (modulation degree) = 0.8				
	Continuous		260		Arms
	$t_{outmax120}$	$t=120s$; S2 operation mode		450	
$t_{outmax2}$	$t=2s$; S2 operation mode		600		Arms
I_{outsp}	protection function	850			A
f_{sw}	Switching frequency	1	8	20	kHz
C_{DC}	DC-link capacitance	11.5	14.4	17.3	mF
$\Delta V_{GND-DClink}$	offset voltage between GND and DC-	-0.3		0.3	V
Environmental Data					
T_a	Temp. of cooling medium	-40		85	°C
T_{env}	Temp. of surrounding air	-40		85	°C
T_{no}	Temp. non-operating	-40		85	°C
IP	ISO 20653			IP6K6 IP00 ⁴	
Altitude	Acceptable operating altitude			4000	m
Driver Data					
V_a	Supply voltage for SKAI3 LV gate driver	12.4	13	13.6	V
t_{TD}	TOP-BOT interlock time	See technical explanation document			
Sensors					
I_{out}	Sensors placed on L1/U and L3/W		2		
T_{DCS}	Sensor placed on power-stage		1		
T_{DCB}	Sensor placed on gate-driver board		1		
Protection Functions					
TOP-BOT interlock		See technical explanation document			
Overcurrent protection (I_{outmax} , single event)					
Gate-driver undervoltage protection					
Signal Interfaces					
Controller to gate-driver (control and sensor connector)		See technical explanation document			
Controller to vehicle (vehicle connector)					
Scaling of signal voltages (V_{jmax} , V_{CC}) and integrated signal conditioning					

¹ No galvanic isolation between power circuit and control circuit. Basic isolation from circuit to heatsink
² Typical nominal battery voltages
³ T_a = temperature of the cooling medium, which could be air (if air cooled) or water (if water cooled)
⁴ R_{thJA} defines the complete thermal resistance from junction of the power switch (T_j) to ambient (T_a)
⁵ typical; may vary depending on control-scheme and application conditions
⁶ IP6K6 for the electronic system under the enclosure; IP00 at power terminals

SKAI 41 A3 MD 10 P Rev. 1.1

Symbol	Conditions	Min.	Typ.	Max.	Unit
Mechanical Data					
Weight			1.95		kg
Height	Height without DC bus terminals		80		mm
Width	Width without signal connectors		150		mm
Length			140		mm
M_1	Mounting torque AC / DC terminals (M8 screw)	8	9	10	Nm
M_2	Mounting torque inverter to vehicle / machine (M8 screw)	18	20	22	Nm

Symbol	Function	Min.	Typ.	Max.	Unit
Width of cable contact on terminals (recommended cable cross section >70mm²)					
DC+	DC BUS "+";	19		22	mm
DC-	DC BUS "-";	19		22	mm
L1 / U	Phase U	19		22	mm
L2 / V	Phase V	19		22	mm
L3 / W	Phase W	19		22	mm

SKAI 41 A3 MD 10 P Rev. 1.1

$P_{cond} [W] = R_\theta \cdot (p_{r1} + p_{r2} \cdot T_j + p_{r3} \cdot T_j^2) \cdot I_{out}^2$		R_θ	0.0024 Ohm
$P_{sw} [W] = f_{sw} \cdot (p_{sw1} + p_{sw2} \cdot I_{out} + p_{sw3} \cdot I_{out}^2) \cdot ((V_{CC} - V_{nom}) \cdot p_v + 1)$		p_{r1}	0.95
$P_{cap} [W] = (R_{cap} \cdot T_{cap} + p_{cap}) \cdot I_{cap}^2$		p_{r2}	0.0035 °C ⁻¹
		p_{r3}	0.000016 °C ⁻²
		p_{sw1}	0.4
		p_{sw2}	0.030 A ⁻¹
		p_{sw3}	0.00003 A ⁻²
		V_{nom}	48 V
		p_v	0.018
		R_{cap} (for $T_{cap} > 20^\circ C$)	-0.0000025 W/(°C*A²)
		p_{cap}	0.00089 W*A²
I_{out} [Arms]	RMS current flowing through DC-link capacitors	$I_{out} = I_{out} \sqrt{\frac{\sqrt{3}m}{\pi} \cdot \frac{9m^2}{\pi} \cdot \cos(\phi)^2 + \frac{\sqrt{3} \cdot m}{4\pi}}$	
T_j [°C]	junction temperature of MOSFET	for high temperature estimation use: $T_j = T_{amb} + 130^\circ C$	
f_{sw} [kHz]	PWM switching frequency		
V_{CC} [V]	DC-supply voltage		
T_{cap} [°C]	Temperature of DC-link capacitors	for high temperature estimation use: $T_{cap} = T_{amb, cap} + 100^\circ C$	
m [-]	Modulation ratio	$m = \sqrt{\frac{V_{outLL}}{V_{CC}}}$	
V_{outLL} [Vrms]	fundamental RMS value of phase-to-phase inverter output voltage		

Simplified powerloss calculation for continuous rotating operation (>10 Hz mech.) using classical Space Vector PWM

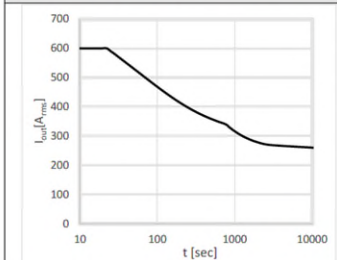


Fig. 1: S2 for $m=0.8$, $\cos(\phi)=0.8$; $V_{CC}=48V$; $f_{sw}=8kHz$; $T_c=40^\circ C$; $T_a=40^\circ C$; $R_{thJA}=0.17\text{ KW}^\circ C$

Annex B - Excerpt of Datasheet of Machine Driven by SKAI3LV-OLEA

Electrical Data					
Number of phases					3
Number of pole pairs					4
Maximal efficiency					>96 %
T/I constant ($I<I_{nom}$)					0.35 Nm/A _{rms}
U/n constant (AC)	rms:	23.8	peak:	33.7	V/(1000rpm)
K_e constant (AC)	rms:	0.057	peak:	0.080	V/(rad*s ⁻¹)
Additional Data					
Weight (w/o cables)					44 kg
Rotor moment of inertia					0.03512 kg*m²
Protection category					IP65 / IP69k
Maximal rotor temperature					150 °C
Allowed ambient temperature					-20 ... 45 ¹⁾ °C
Cooling (medium, flow rate, inlet temperature, pressure)					water/glycol 50/50, 6 l/min, ≤ 60°C, ≤ 0.5 bar
Temperature monitoring					1 x KTY84-130
Type approval					CE, EN 60034
Customs tariff number					8501 5230
Connectors					
Power terminals					3 x M25 cable gland
Signal connectors					M16, 10 Pin
Cooling connectors					2 x ¾" / 19 mm

¹⁾ other range on request

Annex C - Assembly of an Exemplary Customer-designed Controller Board into the SKAI3 LV Standard

View on bottom side of controller board



View on upper side of controller board



Annex C - Assembly of an Exemplary Customer-designed Controller Board into the SKAI3 LV Standard

The vehicle connectors are guided through the dedicated cut-outs in the standard cover



Placing the controller board into the standard cover; upper side of controller board is facing the cover



Annex C - Assembly of an Exemplary Customer-designed Controller Board into the SKAI3 LV Standard

Assembly of an exemplary controller board to the SKAI3 LV Standard

Controller board placement ready for fixation



Controller board is screwed into the standard cover; for required screw type and fastening torque see [\[1\]](#)
Dimension of SKAI3 LV standard version (cooling plate)



Annex C - Assembly of an Exemplary Customer-designed Controller Board into the SKAI3 LV Standard

The standard cover (with the installed controller board) is put on the SKAI3 LV.

Care must be taken when connecting the internal interface between controller board and SKAI3 LV driver board

The standard cover is then re-mounted to the SKAI3 LV plate with the 10 screws
(no picture)

For defined torque see [\[1\], Dimensions of SKAI3 LV Standard Version](#)

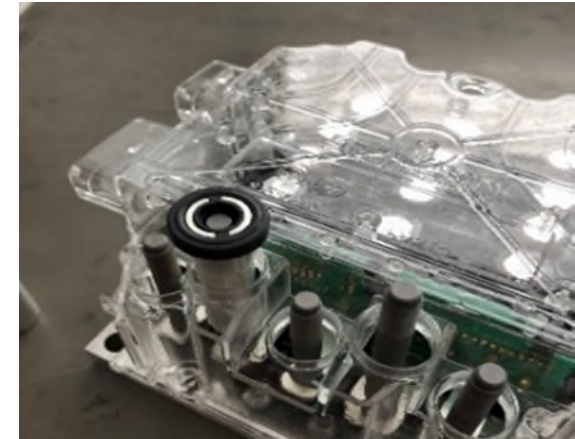


Annex D - Mounting IP seals on power terminals

To ensure the required ingress protection (IP), the seals must be re-assembled on the power terminals.

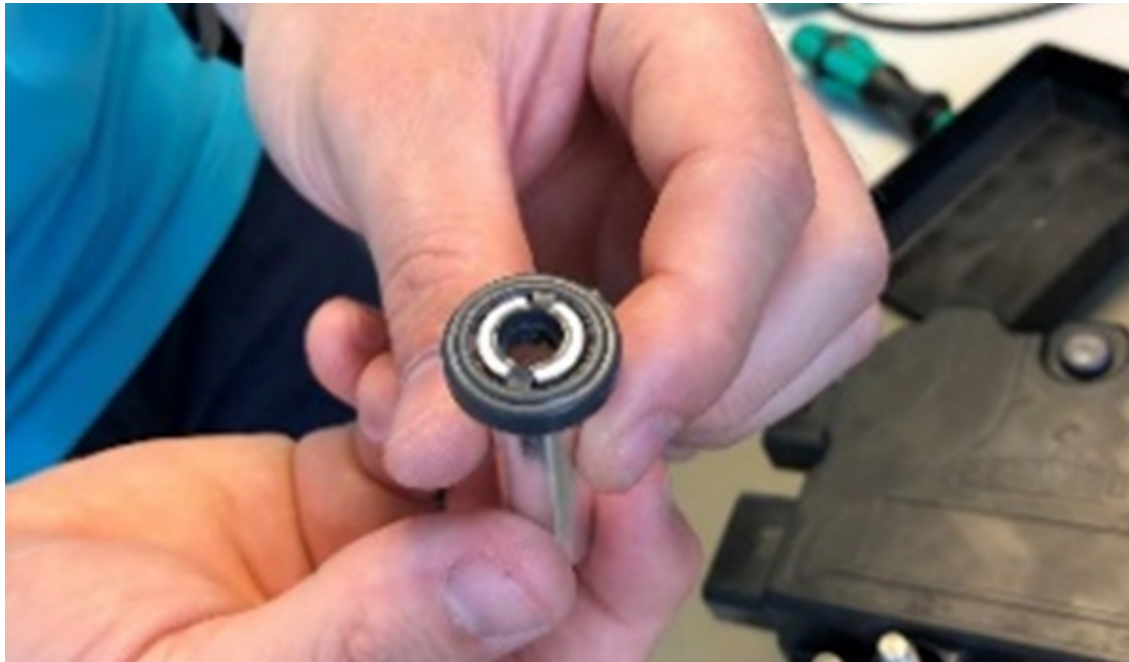
For adequate Annex A - Assembly and avoidance of damage it is strongly recommended to use a suitable tool.

For dimensions of this tool see [\[1\]](#)
Assembly Aid for IP Seals
Installation



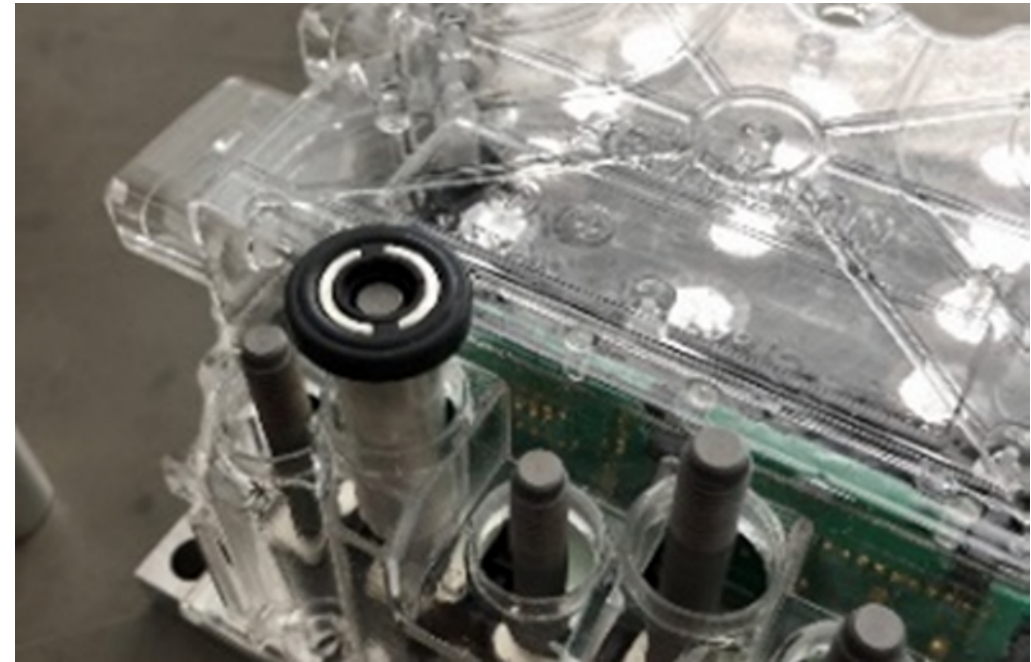
Annex D - Mounting IP seals on power terminals

Place the seal on top of the terminal metal socket. Keep in mind that the seals lug has to be placed inside the sockets groove.



Now place the socket over the appropriate terminals.

Don't press the socket inside the borehole of the SKAI3-LV case.



Annex D - Mounting IP seals on power terminals

Take the assembly tool and press the seal gently over the terminal until you reach the casing of the SKAI3 LV.



Final result should look like this



Liability

IMPORTANT INFORMATION AND WARNINGS

The information in this document may not be considered as guarantee or assurance of product characteristics ("Beschaffenheitsgarantie"). This document describes only the usual characteristics of products to be expected in typical applications, which may still vary depending on the specific application. Therefore, products must be tested for the respective application in advance. Application adjustments may be necessary. The user of SEMIKRON products is responsible for the safety of their applications embedding SEMIKRON products and must take adequate safety measures to prevent the applications from causing a physical injury, fire or other problem if any of SEMIKRON products become faulty. The user is responsible to make sure that the application design is compliant with all applicable laws, regulations, norms and standards. Except as otherwise explicitly approved by SEMIKRON in a written document signed by authorized representatives of SEMIKRON, SEMIKRON products may not be used in any applications where a failure of the product or any consequences of the use thereof can reasonably be expected to result in personal injury. No representation or warranty is given and no liability is assumed with respect to the accuracy, completeness and/or use of any information herein, including without limitation, warranties of non-infringement of intellectual property rights of any third party. SEMIKRON does not assume any liability arising out of the applications or use of any product; neither does it convey any license under its patent rights, copyrights, trade secrets or other intellectual property rights, nor the rights of others. SEMIKRON makes no representation or warranty of non-infringement or alleged non-infringement of intellectual property rights of any third party which may arise from applications. This document supersedes and replaces all information previously supplied and may be superseded by updates. SEMIKRON reserves the right to make changes.



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